



In-season estimates of ad-intact fish returning to the Krassel PIT-array

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Summary

- Current PIT-tag expansions at the Krassel PIT-array (KRS; as of July 17th)¹:
 - 656 natural-origin fish estimated from 118 observed tags and 18% sampling/tagging rate (Tables 1 and 2)
 - Using a 20% tagging rate the estimate is 590.
- KRS was non-operational: 7/10, 7/11, 7/14, 7/15, and 7/16
 - Natural-origin PIT-tag passage during this period is estimated as ≤ 10 tags for an additional 56 natural-origin fish.
- The average passage date of $\geq 75\%$ of the total return across KRS is July 15th (Figure 1).
- End of season natural-origin estimate: 948
 - NOR: $(656 + 56)/.75 = 949$
- Current Lower Granite Estimate of McCall Integrated fish is 465.
 - Estimated at KRS after accounting for SbyC is 443 (95% conversion)
- **Total ad-intact fish at KRS: 1,413**
 - $949 \text{ (NOR)} + 465 \text{ (Integrated)} = 1,414$

Assumptions

- 100% conversion of integrated fish to KRS.
- 100% detection efficiency at KRS
 - In 2021 and 2022 KRS detections efficiency was $\geq 95\%$.
- Tagging rate at LGR of 18%.
 - Only 4 out of 118 NOR tags were marked at LGR during the 29% sampling rate, all others marked at 18%.
- Static interpolation rate of 2 tags during the missed period at KRS when observations are typically decreasing.

¹Scripts to reproduce these summaries are available at "https://github.com/NPTfisheries/sf_salmon_harvest.com".

Table 1: *Current tag expansion rates and estimates for SFSR release groups at KRS.*

Release Year	Group	SbyC	Expansion	n_tags	Estimate
2020	Integrated	RAL	6.710924	1	7
2020	Segregated	RAL	50.028692	1	50
2021	Integrated	RAL	6.995739	51	357
2021	Integrated	RTR	1.000000	21	21
2021	Segregated	RAL	50.761811	45	2,284
2021	Segregated	RTR	1.000000	25	25
2022	Integrated	RAL	8.312441	7	58
2022	Segregated	RAL	39.299292	8	314
2022	Segregated	RTR	1.000000	2	2
2023	NOR		5.555556	118	656

Table 2: *Current SFSR release group estimates at KRS.*

Group	n_tags	Estimate
Integrated	80	443
NOR	118	656
Segregated	81	2,675

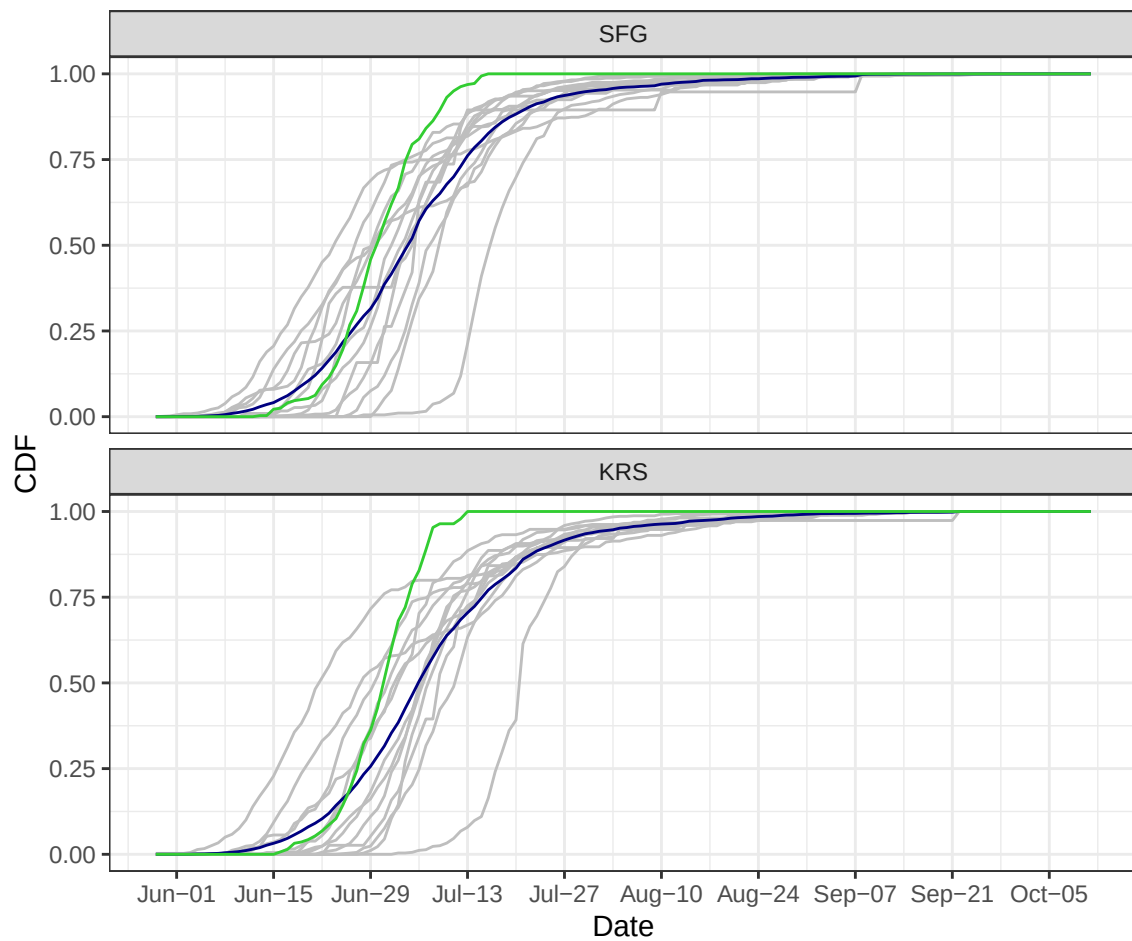


Figure 1: Run-timing of SFSR hatchery (segregated and integrated) and natural origin returns across PIT-arrays. The grey lines indicate individual spawn year returns for 2010-2022, the dark blue line is the average cumulative proportion of returns across the same years, and the green line represents the current 2023 return.